



Project Objectives and Expected Outcomes

SMART Criteria Formulation, Tripartite Pedagogical Outcomes & Course Competencies

Project: AI-Powered Smart Campus Resource Management System	Faculty Coordinator / Mentor: Dr. Rashmi Mothkur (ASAC CSE)
Team Lead: Likith S (2411021061437)	Members: Vinodkumar (2411021061425), Nikhil kumar V (2411021061427), Kalyan kumar T (2411021061421)

1. Core Engineering Objectives

In strict adherence to the project specifications mandated for Alliance University Design Project – I (E1CSA313), the engineering team has defined six core objectives:

- Centralized Campus Management Dashboard:** Develop a unified, responsive campus resource management dashboard integrating administrative, faculty, and student portals for facility booking, scheduling, and operational telemetry.
- Real-Time Utilization Monitoring:** Monitor classroom and laboratory utilization, environmental conditions, and occupancy status in real time using non-intrusive IoT sensor telemetry.
- Machine Learning Demand Prediction:** Predict resource demand, identify peak usage periods, and estimate facility maintenance requirements using machine learning predictive models.
- Automated Timetable & Room Optimization:** Optimize timetable scheduling and room allocation automatically using constraint satisfaction algorithms to eliminate clashes and maximize space utilization.
- Energy Consumption Tracking & Recommendations:** Track electrical energy consumption across campus infrastructure and provide automated, data-driven recommendations for energy conservation and peak-load shaving.
- Administrative Analytics & Decision Support:** Generate comprehensive analytical reports and audit logs for institutional administrators to support strategic infrastructure planning and operational sustainability.

2. SMART Criteria Mapping

Objective	Specific Goal	Measurable KPI	Achievable Strategy	Timeline
Obj 1: Dashboard	Multi-role React web portal	Sub-100ms API latency; 100% mobile-responsive	React 19, Tailwind CSS, Vite architecture	Weeks 1–4
Obj 2: Monitoring	Real-time IoT telemetry ingestion	1,000+ data points; 1 Hz streaming frequency	MQTT broker + WebSocket gateway pipeline	Weeks 3–6
Obj 3: ML Demand	Predict space & equipment demand	RMSE < 0.08, $R^2 > 0.92$, MAE < 5% error	LSTM neural nets & Random Forest regressors	Weeks 5–8
Obj 4: Timetable	Automated conflict-free scheduling	0 hard clashes; +25% seat fill-rate	Constraint Satisfaction Problem (CSP) solver	Weeks 7–10
Obj 5: Energy	Track power & recommend setbacks	18.4% peak grid reduction; 15% kWh savings	Dynamic chiller setback rules & BESS dispatch	Weeks 8–11
Obj 6: Analytics	Executive audit reports	1-click PDF/CSV export; 100% audit logging	Aggregated PostgreSQL reporting queries	Weeks 10–12

3. Tripartite Project Outcomes (Process Manual Section 1.4)

- Cognitive Outcomes:** Formulated multi-variable mathematical optimization models balancing space demand, instructor preferences, and chiller thermal inertia; derived thermodynamic building differential equations; evaluated alternative system topologies using Pugh decision matrices.
- Practical & Technical Outcomes:** Built a production-ready centralized dashboard with integrated 3D WebGL Digital Twin; engineered low-latency IoT state simulation streaming 1,000+ telemetry attributes; validated 18.4% grid peak demand reduction.
- Professional & Soft Outcomes:** Authored comprehensive IEEE technical documentation; managed 12 sequential weekly review logs signed by Dr. Rashmi Mothkur; maintained strict ethical privacy standards eliminating biometric facial surveillance.

4. Mapped Curricular Competencies

The project directly aligns with five upper-level computer science subjects: **Artificial Intelligence** (CSP formulation, state-space tree search), **Machine Learning** (time-series forecasting, hyperparameter tuning), **Cloud Computing** (stateless microservices, containerization with Docker), **Mobile Application Development** (responsive viewport scaling, touch gesture controllers), and **Internet of Things** (sensor telemetry, payload serialization, MQTT/WebSocket protocols).